

Computational workflow

Shared field conventions; separate state equations

1 Set parameters

Pulse parameters and units
Momentum grid p/m
Tail extent L/τ
Solver tolerances

4 Validate each run

Exact Sauter benchmark
Final norm and invariant
Dirac-QKE agreement
Tail-window scan

2 Define the field

Sauter pulse or pulse train
 $E(t)$, $A(t)$, finite window

5 Save datasets

Spectrum and yield CSV
Tail and $p-L/\tau$ map CSV
Run metadata JSON

3 Solve each mode

Dirac spinor: DOP853
QKE state (f, u, v) : DOP853
Separate integrations

6 Generate and build

Python/Matplotlib figures
LaTeX: two-pass CAS PDF
Package checksums

reproduce.py · make_reproducibility_figures.py

build_manuscript.sh · verify_checksums.py